

Campus Neighbour Project Proposal
Campus Second-Hand Trading Platform

Project information	Details
Group number	[To be filled]
Programme of study	[To be filled]
Academic supervisor	[To be filled]
Proposal submission deadline	25 September 2026, 23:59 CST
Project start and end dates	[To be filled]
Group members and student IDs	1. YUHAN JIAO 50106173 2. ZIYE JIANG 50106131 3. LU LIU 50106174 4. RUIYU GONG 50106169 5. Han Huang 50106171 6. YIAYI WU 50106143 7. JIAYI ZHENG 50106140 8. RUIJIE LI 50106150 9. JUNYU PAN 50106130

Proposed project focus

A campus second-hand trading platform built around local discovery, private communication, in-person handover and transaction feedback. Later iterations will improve the same trading workflow through direct barter and convenient buyer-seller chat.

1 Title

Design and Implementation of Campus Neighbour

A campus second-hand trading platform

2 Problem

The project addresses three problems in campus second-hand trading. Fragmented information: listings and conversations are scattered across group chats and separate channels, making items difficult to find and compare. Poor transaction status visibility: buyers may not know whether an item is available, reserved or sold, while informal agreements leave handover and completion unclear. Lack of campus-specific discovery: searches need to account for campus buildings, courses and textbook editions so students can identify suitable items and arrange nearby collection. A shared campus marketplace will bring this information and the trading process together, supporting local reuse [1].

3 Solution

3.1 Core platform

We propose a responsive web application for campus second-hand trading. A seller publishes an item, a buyer browses or searches, both parties communicate through private chat, they agree an in-person handover, the buyer confirms receipt on a phone, and both parties can leave feedback. The platform will support student and administrator accounts, listing moderation and notifications.

Clicking "Pay" opens a demonstration page displaying "Payment methods are currently unavailable", with a return option. No payment provider is connected, no funds are transferred and no payment success is recorded. Delivery and escrow remain outside scope.

3.2 Campus features and iteration

Listings will include photographs, a description, condition, price, category and campus location. Textbook listings may also include the course, author and edition. Buyers can filter by item, price, building and course. Listing status will show whether an item is available, reserved or sold. A seller can reserve an item for one buyer; cancellation releases the reservation. Reviews will be available to participants after completion. Graduation and new-semester collections can group relevant listings.

Buyer and seller profiles will provide a short introduction with fields for school or faculty, degree programme and year of study. For course-related exchanges, users can add the relevant course, its instructor and a brief description of the course. This background will help participants understand one another and discuss whether a textbook matches their course requirements.

Development will deepen this trading workflow in stages. The first release will establish publishing, filtering, chat, handover confirmation and reviews. A later iteration will add direct one-to-one barter: each user states what they offer and what they want, both accept the match, and the existing reservation and handover process is reused. Complex multi-person swaps and cash adjustments are outside scope.

3.3 Real-time chat and system design

The buyer-seller real-time chat window will offer four choices: Ask about price, Ask about item condition and availability, Ask about the meeting location, and Free chat. The first three choices prepare a preset question for the user to send to the other participant. Free chat allows either participant to compose a message of up to 20 characters; multiple messages may be sent. Replies come from the other participant, not an AI assistant. Preset questions and free-text messages share the same conversation history, notifications and access controls.

The proposed system consists of a browser interface, an application server and a relational database. Main records include users, buildings, courses, listings, conversations, transactions, barter preferences and reviews. Access controls will protect private messages and transaction actions. Item reservation and receipt confirmation will be recorded as controlled state changes to prevent conflicting transactions.

3.4 Future work

Future work may extend the platform with campus maps, elective-course reviews linked to textbooks, life guides and food information. Advanced routing, personalised recommendation, payment custody, escrow, delivery management, merchant self-service and multi-person barter are also reserved for later development. These functions are not required for the proposed proof of concept.

4 Objectives

O1 Requirements and design

Define the campus trading requirements, use cases, interface prototype, architecture, data model and acceptance tests.

O2 Core trading workflow

Implement accounts, listings, filters, private chat, reservation, in-person handover confirmation and transaction-based reviews.

O3 Campus enhancements

Implement buyer and seller profiles, building and course filters, seasonal collections and direct one-to-one barter. Provide preset question buttons and short free-text messages within buyer-seller real-time chat.

O4 Testing and delivery

Test the main workflows, permissions and transaction states, resolve critical defects, and deliver reproducible software and course documentation.

Evaluation approach

Evaluation will use agreed acceptance tests and student task sessions to check the trading flow, listing filters and transaction states. Chat tests will cover the three question buttons, the 20-character free-message limit, replies from the other participant, access control, message history and failed-send recovery. Final targets will be confirmed after requirements and sample data are available.

5 Benefits

A practical campus trading process

Students can find nearby goods, communicate within the relevant listing and complete a transaction without logistics. Course and edition fields make textbook searches more precise, while building filters support convenient handover.

Reservation and confirmation reduce uncertainty about availability, and reviews are linked to completed transactions.

Progressive improvement within one clear scope

The project concentrates development and testing on one business process. Barter, seasonal collections and convenient real-time chat improve the same marketplace.

Preset questions and short free-text messages help buyers and sellers communicate about the item and arrange a handover. This gives the team a clear minimum release while preserving room for technical and product innovation.

Reuse and sustainability

Reselling and exchanging textbooks and everyday items can extend their useful life and reduce unnecessary disposal. Campus-based discovery and seasonal collections will help students pass usable belongings to others who need them. The project aims to make reuse convenient and support more sustainable consumption within the campus community.

6 Timeline

The following dates follow the course schedule. All submission deadlines are at 23:59 CST (China Standard Time, UTC+8). Supervisor allocations will be announced by 18 September; no announcement time is specified.

Activity or deliverable	Deadline in 2026	Weight
Join a group on MyAberdeen	Monday, 14 September	—
Supervisor allocations announced	Friday, 18 September	—
Project proposal	Friday, 25 September	0%
Project Update 1	Monday, 12 October	0%
Project Update 2	Monday, 2 November	0%
Project Update 3	Monday, 23 November	0%
Project Update 4	Monday, 7 December	0%
Technical Report	Monday, 14 December	50%
Proof-of-Concept Software	Monday, 14 December	30%
Presentation	Monday, 14 December	20%

The proposal is mandatory although unmarked; missing or late submission may result in a penalty to the overall group project mark. At least three of the four project updates are mandatory; submitting fewer may also incur a penalty. Update forms are

available under Course Assessment in the Group Project folder on MyAberdeen.
See the project description and submission requirements for full details.

7 Action plan

Internal task assignments, deadlines and progress values are to be filled by the group in line with the course dates above.

Objective O1 Requirements and design

Action	Assigned to	Deadline	Progress
Collect user needs and agree the project scope.	[To be filled]	[To be filled]	[To be filled]
Prepare use cases, prototype, architecture and data model.	[To be filled]	[To be filled]	[To be filled]

Objective O2 Core trading workflow

Action	Assigned to	Deadline	Progress
Build accounts, listings, filters and private chat.	[To be filled]	[To be filled]	[To be filled]
Build reservation, handover confirmation and reviews.	[To be filled]	[To be filled]	[To be filled]

Objective O3 Campus enhancements

Action	Assigned to	Deadline	Progress
Add user profiles, campus-specific fields, seasonal collections and direct barter.	[To be filled]	[To be filled]	[To be filled]
Implement and test preset question buttons and short free-text messages in buyer-seller chat.	[To be filled]	[To be filled]	[To be filled]

Objective O4 Testing and delivery

Action	Assigned to	Deadline	Progress
Test permissions, trading states, search and concurrency.	[To be filled]	[To be filled]	[To be filled]
Run student tasks, resolve defects and repeat affected tests.	[To be filled]	[To be filled]	[To be filled]
Prepare the release, report, manual and presentation.	[To be filled]	[To be filled]	[To be filled]

References

[1] Ministry of Commerce and eight other departments (2026). Notice on Implementing the Green Consumption Promotion Action. 4 January. Sections 9 and 12. [Official policy text](#)

Source accessed 15 September 2026. The English title is a descriptive translation of the Chinese original.